

Coastal Hydrodynamics IV

Course : CE60222

Dr. Mohammad Saud Afzal

Associate professor

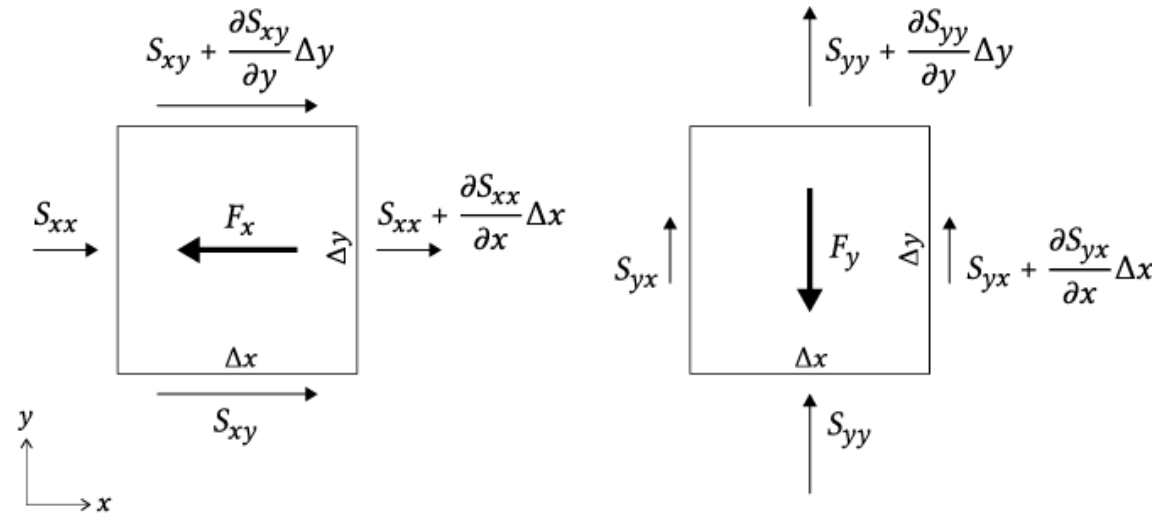
Civil Engineering Department



Contents

- Wave-induced forces
- Cross-shore balance: wave set-up and set-down
- Cross-shore mass balance (alongshore uniform coast)
- Cross-shore momentum balance (alongshore uniform coast)
- Wave set-up and set-down
- Wave set-down in the case of normally incident waves
- Wave set-up in the case of normally incident waves

Wave-induced forces

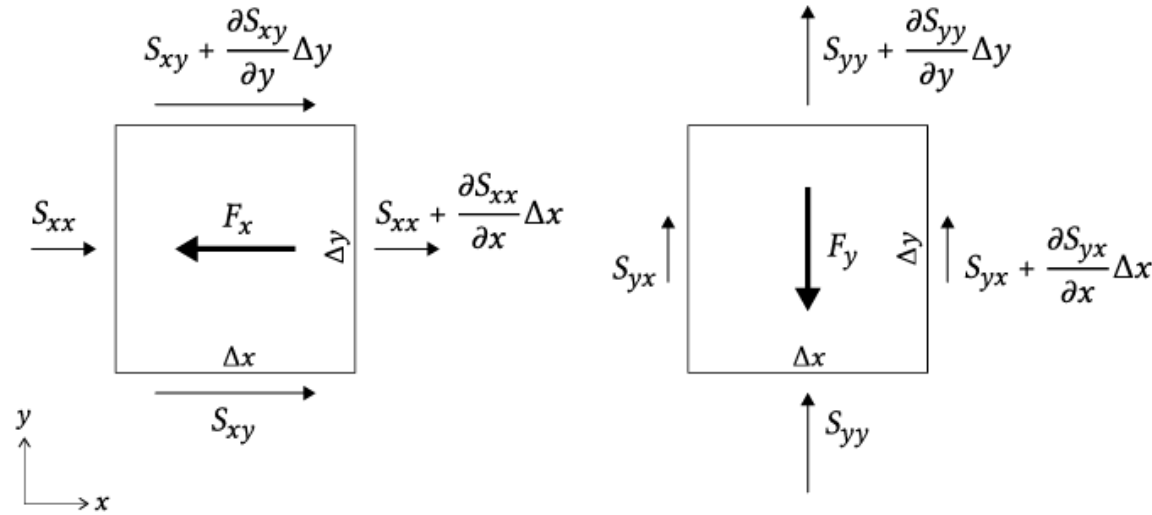


- **Horizontal gradients** in the radiation stresses give rise to a **net wave-induced force** on the water in a particular direction. The net force in the **x -direction** is described by

$$F_x = - \left(\frac{\partial S_{xx}}{\partial x} + \frac{\partial S_{xy}}{\partial y} \right)$$

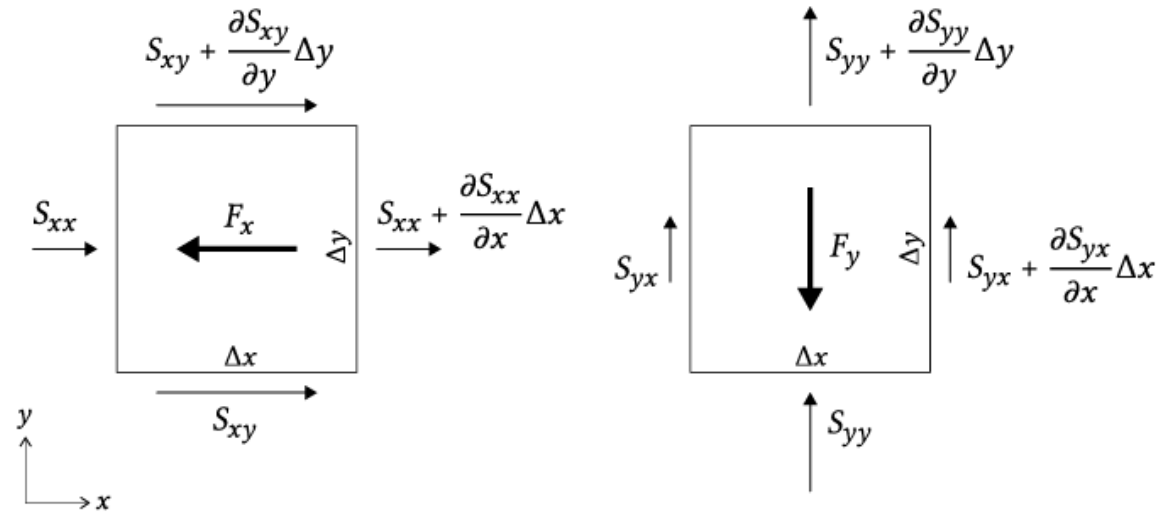
Where: $\frac{\partial S_{xx}}{\partial x}$ = the effect of variations in the x -directed **radiation normal stresses**

and $\frac{\partial S_{xy}}{\partial y}$ = the effect of variations in the y -direction of the x directed **radiation shear stress**.



- In **coastal engineering practice**, it is common to work with **alongshore** and **cross-shore** orientated axes.
- **The y-axis** is defined **parallel** to the shoreline, while the **x-axis** is **perpendicular** to the shoreline. F_x is thus the force in the **cross-shore direction**.
- The **shear term** is zero for an **alongshore uniform coastline** (no gradients in the y-direction). Then we have:

$$F_x = - \frac{\partial S_{xx}}{\partial x}$$

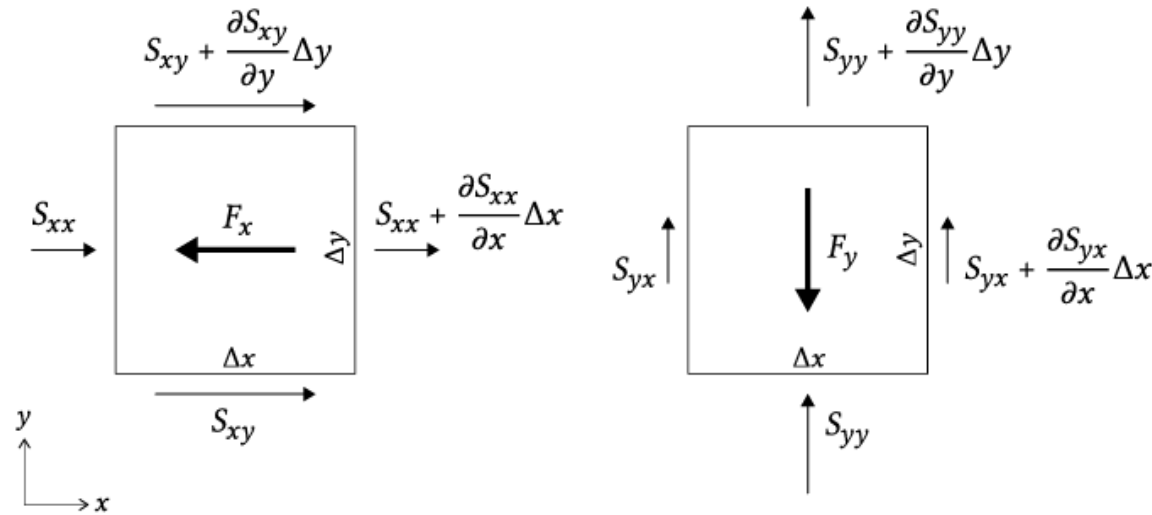


- In the **alongshore direction** the force is:

$$F_y = - \left(\frac{\partial S_{yy}}{\partial y} + \frac{\partial S_{yx}}{\partial x} \right)$$

- The **first term** can be **non-zero** if gradients in wave height occur **along the coast**.
- For an **alongshore uniform coast** this term is **zero** such that Equation reduces to:

$$F_y = - \frac{dS_{yx}}{dx}$$



- Here S_{yx} – **the radiation shear stress** on a plane **perpendicular** to the **x -axis** – represents the transport of **y -momentum** in the x -direction.
- A **cross-shore gradient** here gives a **net force** in the y -direction.
- Variations in the **radiation stresses** occur due to changes in **n, E or θ** .
- Further **offshore**, the **wave-induced forces** are relatively **small**, but in the **nearshore** zone **large forces** occur due to **wave transformation** as a result of **large gradients** in water depth.

Cross-shore balance: wave set-up and set-down

Wave forces have an effect on the **mean flow**; they induce:

- **Mean water level** variations (**set-down**, **set-up**), and
- **Mean currents** (**alongshore current** in the case of waves obliquely **approaching the shore**).
- First we consider the **wave force** in the **cross-shore direction**:

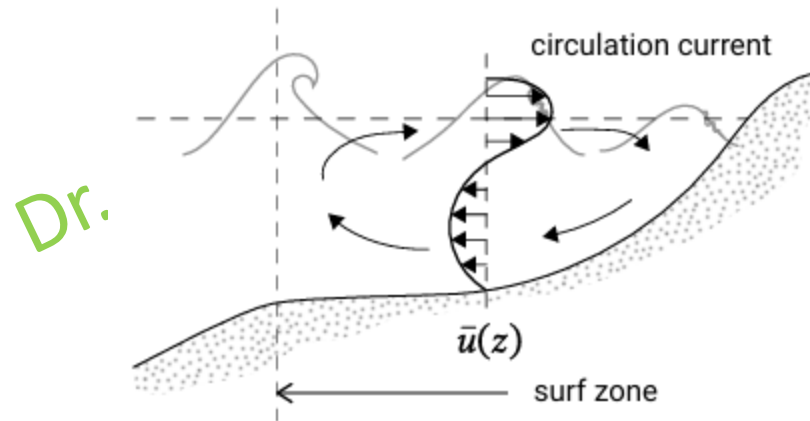
$$F_x = - \left(\frac{\partial S_{xx}}{\partial x} + \frac{\partial S_{xy}}{\partial y} \right)$$

- For **illustration purposes**, we consider the simplified situation of a **long-crested wave** normally incident to **straight and parallel** depth contours (**parallel to the y-axis**, $\theta = \varphi = 0$).
- This means that there are **no gradients** in the **alongshore direction**: the situation is **alongshore uniform** (all y-derivatives are zero) and the wave force reduces to:

$$F_x = - \frac{\partial S_{xx}}{\partial x}$$

Cross-shore mass balance (alongshore uniform coast)

- In a **stationary case**, the **cross-shore current** averaged over the entire water column must be **zero** (since water **neither piles up higher** and higher against the coast **nor flows towards deeper** water).
- **At every point** in the **cross-shore profile**, the onshore- directed mass flux near the water surface is therefore compensated by an offshore-directed **return current** at lower elevations, such that the **net depth-averaged** flow through each **cross-section is zero** (see Figure).



- The **offshore-directed depth-mean velocity** under the wave trough level can be found from Equation.

$$U_{below\ trough} = -\frac{q_{drift,x}}{\rho h} = -\frac{q_{drift} \cos \theta}{\rho h}$$

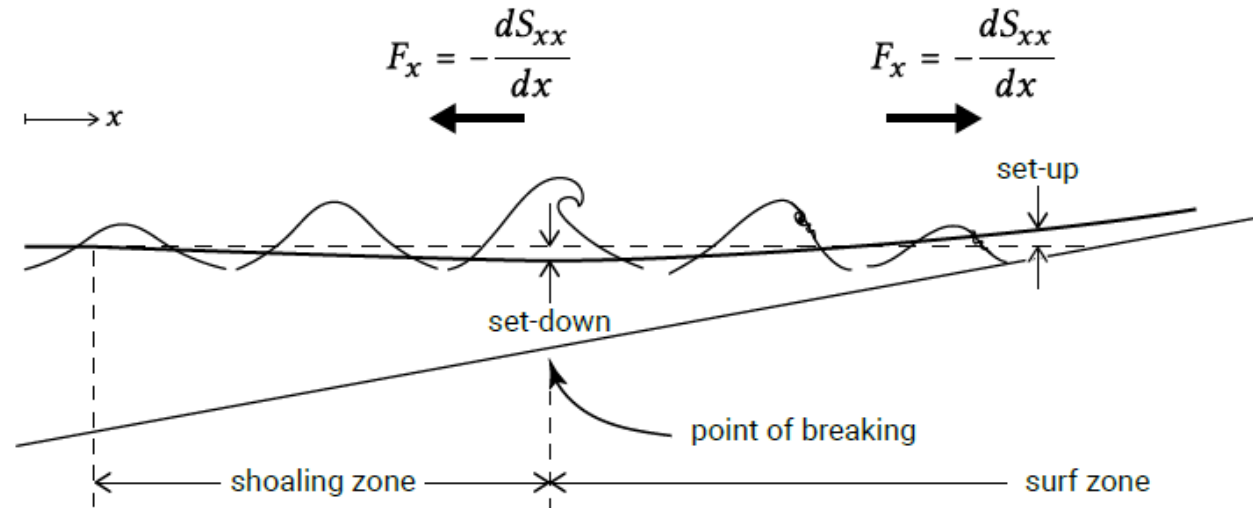
Cross-shore momentum balance (alongshore uniform coast)

- The **wave force** is determined by the **cross-shore gradient** of S_{xx} .
- The magnitude of the **radiation stress** in the **wave propagation** direction.

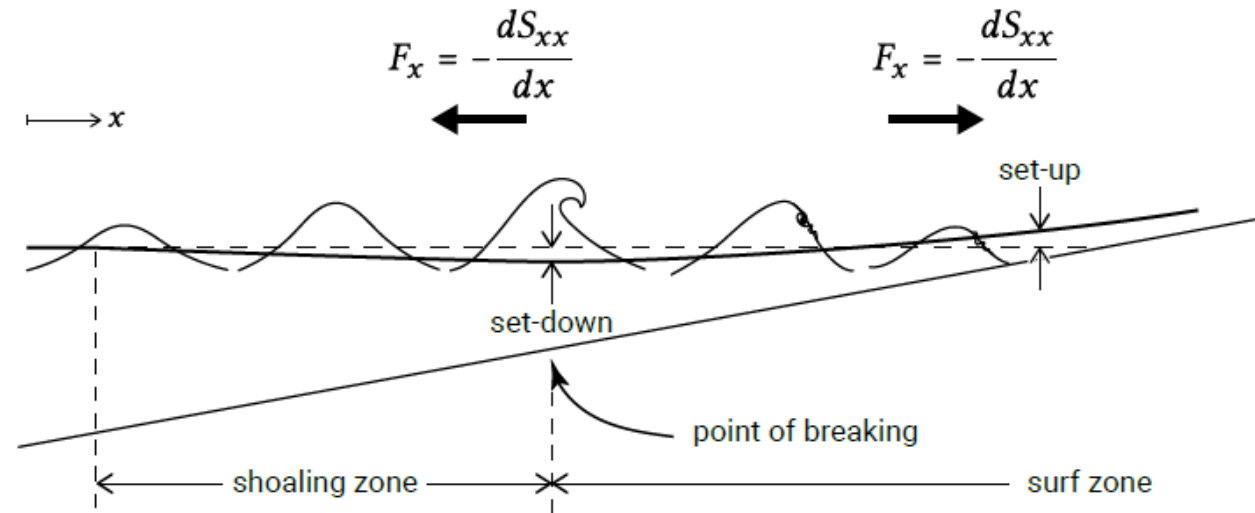
$$S_{xx} = (2n - 1/2)E$$

- Depends on **the wave height, the water depth and the wavelength**.
- Based on **energy conservation**, the **wave energy** (wave heights) tends to **increase** when the waves approach the **surf zone** (after a small initial decrease).
- Since **n increases** in intermediate water depths, **S_{xx} increases** in the shoaling region.
- Further **offshore** in really **deep water**, the **radiation stress** S_{xx} is constant.

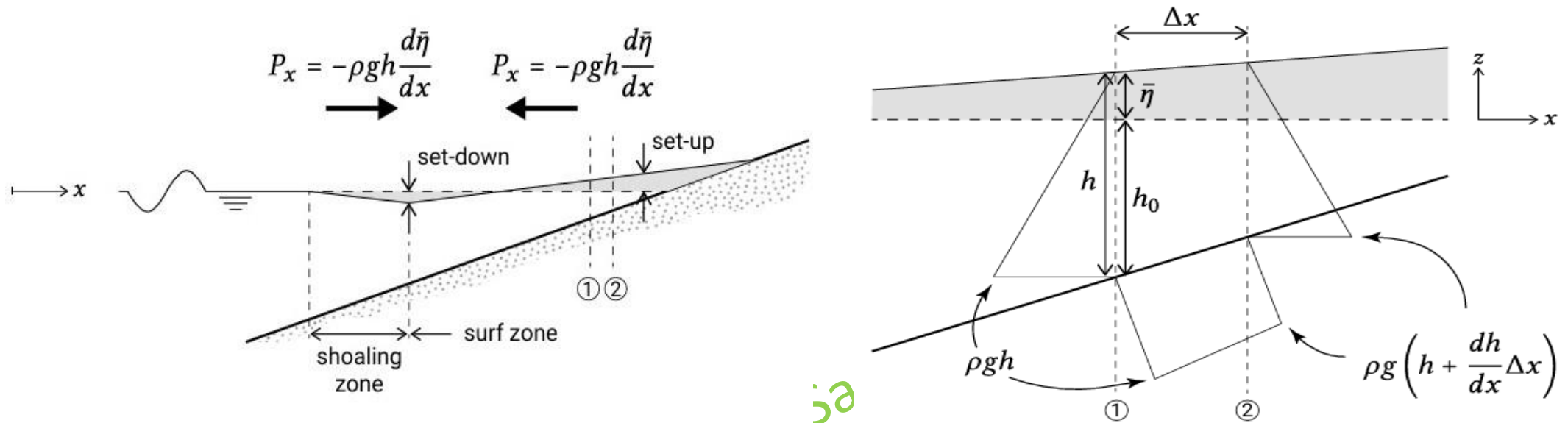
Wave set-up and set-down



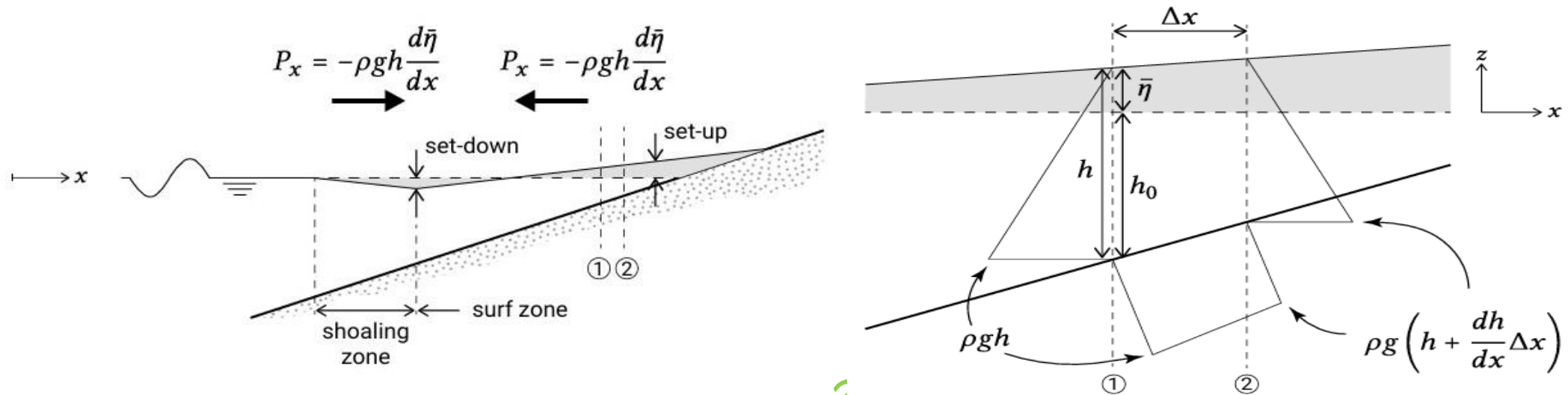
- An **increasing** S_{xx} in the landward direction means that, at a water column, a **resulting force** due to **radiation stresses** is acting in the **seaward direction**.
- A (small) **difference** in water level at **both sides** of the water column (**lower towards the coast**) ensures that **equilibrium** of forces is achieved again.
- This **phenomenon** is called **wave set-down**, which means that **outside the breaker zone** in intermediate water depths, the water level at the **landward side** of a water column is a bit **lower** than at the **seaward side**.



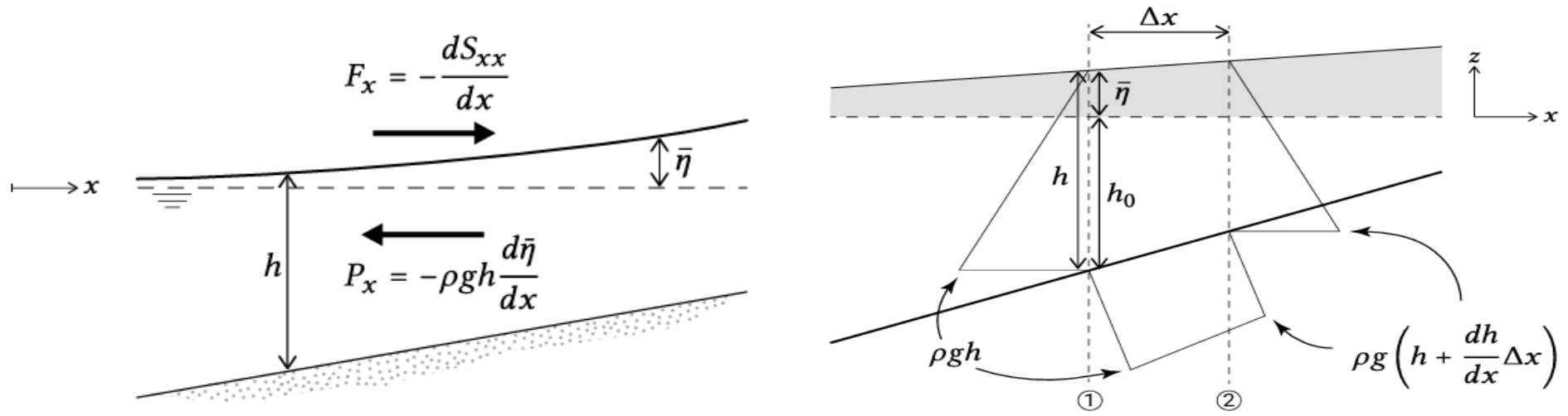
- **Inside the surf zone**, the magnitude of S_{xx} **decreases** rapidly due to **wave-breaking** while moving towards the waterline.
- The **decrease** in S_{xx} is **equivalent** to a force in the **landward direction**.
- To achieve **equilibrium again**, the water level at the **landward side** of the column should be **higher** than at the seaward side creating a seaward-directed pressure force. This **phenomenon** is called **wave set-up**.



- The **control volume** for a simple cross-shore **momentum balance** between two **locations 1 and 2**.
- The **radiation stress** is $S_{xx,1}$ at **location 1** and $S_{xx,2}$ at **location 2**.
- The **triangles** and **quadrangle** indicate the **pressure forces**.
- If **locations 1 and 2** are in the **surf zone**, the onshore directed wave force (onshore since $S_{xx,1} > S_{xx,2}$) is **balanced** by a **net offshore pressure force** (through a raising of the water level towards the coast)



- The above figure illustrates the **cross-shore balance** of **momentum** between two (arbitrary) **points 1 and 2**.
- The **net pressure force** (per unit alongshore distance) is $P_x \Delta x = -\rho g h \frac{d\bar{\eta}}{dx} \Delta x$. It consists of:
 - The hydrostatic force $1/2 \rho g h^2$ at point 1; minus
 - The hydrostatic force $1/2 \rho g (h + \frac{dh}{dx} \Delta x)^2 \approx 1/2 \rho g h^2 + \rho g h \frac{dh}{dx} \Delta x$ at point 2; minus
 - The horizontal component $\approx \rho g h (\frac{dh}{dx} \Delta x)$ of the hydrostatic force along the bottom.



- The **force balance** between **wave force** and **pressure force** reads $F_x + P_x = 0$.
- The **equilibrium** between the **radiation stress gradient** and **pressure term** due to the **water level slope** (see Figure) yields the following **first-order differential equation**:

$$F_x = -\frac{dS_{xx}}{dx} = \rho g h \frac{d\bar{\eta}}{dx} = \rho g (h_0 + \bar{\eta}) \frac{d\bar{\eta}}{dx}$$

where: x = co-ordinate axis pointing from land to sea in m

h_0 = still-water depth at point x (in absence of waves) in m

η = wave-induced water level set-up at point x in m

- **Equation** is the **depth-integrated cross-shore momentum** equation for a **stationary alongshore** uniform situation.
- This balance in the **cross-shore** direction holds both for **obliquely** and **normally incident waves**.
- We have not included **bottom friction** in this equation, because it is generally assumed to be an order of **magnitude smaller** than the other terms in the equation.
- Furthermore, the **Coriolis term** is assumed to be **small** compared to the **wave forcing**.
- The equation is **valid inside and outside** the **surf zone** under the conditions listed above.
- **Inside (outside)** the surf zone the term dS_{xx}/dx is **negative (positive)** yielding a **positive (negative)** water level gradient ($d\bar{\eta}/dx$).
- Therefore, the **still water surface** becomes **lower** in the **shoaling region (set-down)**.

- Moving inside the **surf zone** towards the waterline, the still water surface becomes **higher (set-up)**.
- In order to calculate the **wave set-up (and set-down)**, we need to assess the **radiation stress** terms **inside and outside** the **surf zone**.
- **first-order differential equation**

$$F_x = -\frac{dS_{xx}}{dx} = \rho g h \frac{d\bar{\eta}}{dx} = \rho g (h_0 + \bar{\eta}) \frac{d\bar{\eta}}{dx}$$

- **Integrate** above equation with **respect to x** .
- With Equation $S_{xx} = \left(n - \frac{1}{2} + n \cos^2 \theta\right) E$ we can write **above** equation as:

$$-\frac{d}{dx} \left[\left(n - \frac{1}{2} + n \cos^2 \theta \right) E \right] = \rho g h \frac{d\bar{\eta}}{dx}$$

- The **spatial wave energy** variation must be solved from the **conservation of energy**,

Equation,

$$\frac{d}{dx}(Ec_g \cos \theta) + \frac{d}{dx}(Ec_g \cos \theta) = -D_f - D_w$$

Under the **present assumptions** equation reduces to

$$\frac{d}{dx}(Ec_g \cos \theta) = -D_w$$

Wave set-down in the case of normally incident waves

- For waves **normally incident** to an **alongshore uniform coast** ($\theta = \varphi = 0$) and under the assumption of **shallow water** ($n = 1$), the **momentum balance Equation** can be written as:

$$-\frac{d}{dx} \left(\frac{3}{2} E \right) = \rho g h \frac{d\bar{\eta}}{dx} \quad (\text{i})$$

- Energy balance** for a normally incident **shoaling wave**:

$$\frac{H_2}{H_1} = \sqrt{\frac{c_{g,1}}{c_{g,2}}} \quad (\text{ii})$$

- Longuet-Higgins and Stewart (1962)** derived a formula for the **set down** for **regular waves** by integration of Equation (i) using the energy balance of shoaling waves Equation (ii).
- The **integration constant** is determined from $\eta = 0$ in **deep water**. It can be shown that, in the shallow-water approximation, the **set-down** then equals:

$$\bar{\eta} = -\frac{1}{16} \frac{H^2}{h}$$

- For **waves propagating** without dissipation towards the shore, the **wave height increases** and the **water depth decreases**.
- Hence, the **set-down gradually increases** in magnitude to a **maximum** just **outside the breaker zone**, at the point where the waves start breaking. At the **breaker point the set-down** equals:

$$\overline{\eta}_b = -\frac{1}{16} \frac{H_b^2}{h_b} = -\frac{1}{16} \gamma H_b$$

Where,

$\overline{\eta}_b$ = water level change at the point of wave-breaking m

γ = wave-breaking index (H_b/h_b)

H_b = wave height at point of breaking m

h_b = still water depth at point of breaking m

Thus, with γ put **equal to 0.8** the **set-down** at the point of **breaking is 4 %** of the **local water depth**.

Wave set-up in the case of normally incident waves

- Inside the **surf zone**, the **dissipation** due to **wave-breaking** needs to be included in the **energy balance**.
- An **analytical equation** for **wave set-up** is derived using a very **simple dissipation model** and assuming **normally incident waves**.
- The derivation considers **equilibrium of forces** for the **entire breaker** or **surf zone**.
- A simple model for the **energy dissipation** due to **wave-breaking** assumes that the **wave height everywhere** in the **surf zone** is **proportional** to the **local water depth**, everywhere in the surf zone:

$$H = \gamma h$$

- If we further use the **shallow-water approximation** ($n = 1$), then it follows that $S_{xx} = 3/2E$ is **decreasing** from the **breaker line** towards the **coastline**.
- Following similar reasoning as in the previous section, this **causes landward-directed resulting forces acting on a water column**.

- This will be balanced by a water level increase (in the landward direction) over the water column.
- If we substitute $S_{xx} = 3/2E = 3/16\rho gH^2 = 3/16\rho g\gamma^2 h^2$ in Equation $-\frac{d}{dx}\left(\frac{3}{2}E\right) = \rho gh \frac{d\bar{\eta}}{dx}$,

we find the following balance:

$$-\frac{d}{dx} [3/16\gamma^2 h^2] = h \frac{d\bar{\eta}}{dx}$$

and thus:

$$\frac{d\bar{\eta}}{dx} = -3/8\gamma^2 \frac{dh}{dx}$$

Substituting $h = h_0 + \eta$ in above equation gives:

$$\frac{d\bar{\eta}}{dx} = -\frac{3/8\gamma^2}{(1 + 3/8\gamma^2)} \frac{dh_0}{dx}$$

- Hence, because the water depth decreases in the surf zone ($dh_0/dx < 0$), the set-up increases ($d\bar{\eta}/dx > 0$).
- With a constant bottom slope and a constant breaker index, the water level slope inside the breaker zone is constant.

$$\frac{d\bar{\eta}}{dx} = -\frac{3/8\gamma^2}{(1 + 3/8\gamma^2)} \frac{dh_0}{dx}$$

- Above equation can be integrated from the breaker point to the water line to get the wave set-up throughout the surf zone (using $\bar{\eta} = \bar{\eta}_b$ for $h = h_b$ to find the integration constant C):

$$\bar{\eta} = -\frac{3}{8}\gamma^2 h + C = \bar{\eta}_b + 3/8\gamma^2(h_b - h)$$

- At the shoreline, the water depth h is zero and we find for the set-up at the water line:

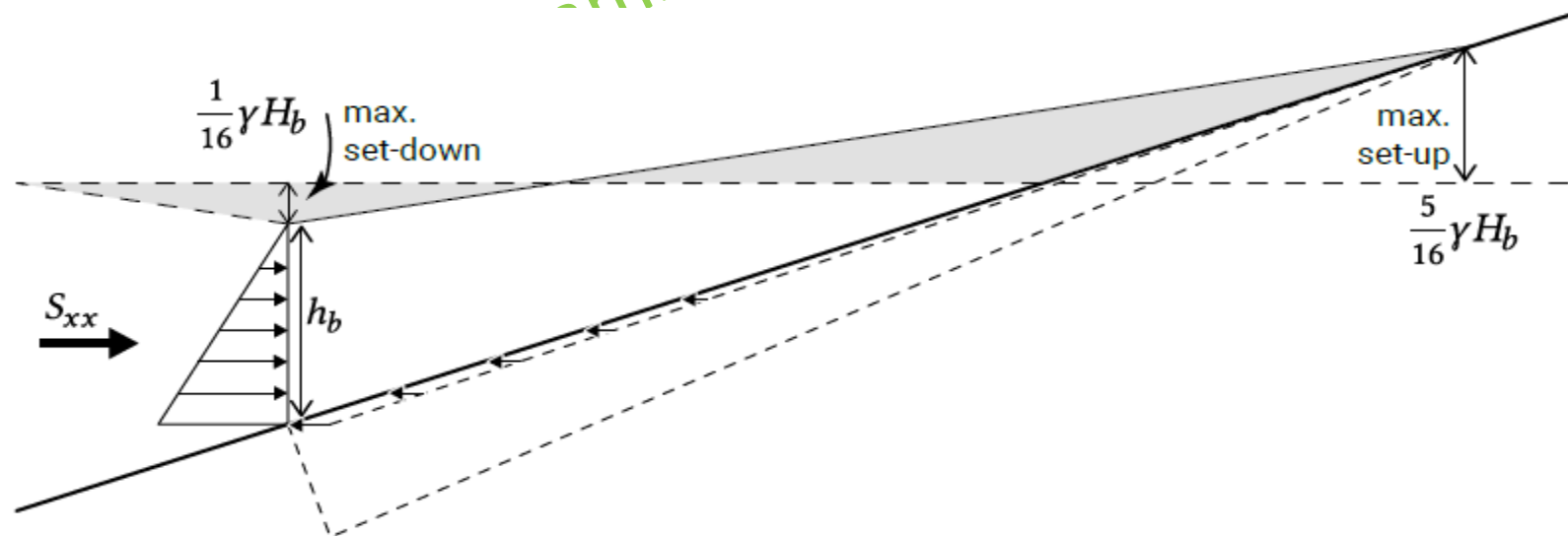
$$\bar{\eta}_{\text{shore}} = \bar{\eta}_b + 3/8\gamma^2 h_b = \bar{\eta}_b + 3/8\gamma H_b$$

- The water level difference $\Delta\bar{\eta}$ between the breaker line and the point of maximum water level rise (wave set-up) equals:

$$\Delta\bar{\eta} = \frac{3}{8}\gamma H_b$$

With $\gamma = 0.8$ this amounts to $0.3H_b$.

- Since the maximum set-down at the breaker line is $1/16\gamma H_b$, the maximum wave set-up relative to the still water level is $5/16\gamma H_b$.



Problems

A monochromatic wave with a height of $H_0=3$ m and a period of $T=10$ s propagates toward a shoreline with a uniform slope of 1:50 . Assuming the wave breaks at a depth of $h_b=5$ m, calculate the wave set-up and wave set-down at the breaking point and at the shoreline.

Solution : Given $H_0=3$ m , $T=10$ s , $h_b=5$ m

$$L_0=1.56 \times T^2=1.56 \times 10^2=156 \text{ m}$$

Wave height at Breaking Point

$$\frac{H_b}{H_0} = 0.5 \left(\frac{L_0}{H_0} \right)^{\frac{1}{5}}$$
$$H_b = H_0 \times 0.5 \left(\frac{L_0}{H_0} \right)^{\frac{1}{5}} = 3 \times 0.5 \left(\frac{156}{3} \right)^{\frac{1}{5}} = 3.31 \text{ m}$$

Wave Set-down at Breaking Point, $\eta_{setdown} = \frac{1}{16} \gamma H_b = \frac{1}{16} \frac{H_b^2}{h_b}$ [as $\gamma = \frac{H_b}{h_b}$]

$$\eta_{setdown} = \frac{3.31 \times 3.31}{16 \times 5} = 0.136 \text{ m}$$

Wave Setup at Shoreline, $\eta_{setup} = \frac{5}{16} \gamma H_b = \frac{5}{16} \frac{H_b^2}{h_b}$ [as $\gamma = \frac{H_b}{h_b}$]

$$\eta_{setup} = \frac{5 \times 3.31 \times 3.31}{16 \times 5} = 0.685\text{m}$$

Dr. Mohammad Saud Afzal

Problems

3 m high wave in deep waters is directed perpendicular to the coast with a seabed slope of 1 in 100. If the wave period is 10 s, determine

- I. wave breaking characteristics,
- II. wave set-down till the breaker-line and
- III. wave setup in the surf zone. Assume breaker index = 0.9.

Calculate breaker depth, wave heights corresponding to different water depths considering shoaling alone.

Calculate set-down

$$\bar{\eta}_{sd} = -\frac{1}{8} \frac{kH^2}{\sinh(2kh)}$$

Calculate setup

$$\bar{\eta}_{su} = \frac{h_b - h}{1 + (8/3\gamma^2)}$$

Solution: Given $H_0=3 \text{ m}$, $T=10 \text{ s}$, $\gamma = 0.9$

$$L_0=1.56 \times T^2=1.56 \times 10^2=156 \text{ m}$$

Wave height at Breaking Point

$$\frac{H_b}{H_0} = 0.5 \left(\frac{L_0}{H_0} \right)^{\frac{1}{5}}$$

$$H_b = H_0 \times 0.5 \left(\frac{L_0}{H_0} \right)^{\frac{1}{5}} = 3 \times 0.5 \left(\frac{156}{3} \right)^{\frac{1}{5}} = 3.31 \text{ m}$$

Breaker index

$$\gamma = 0.9 = \frac{H_b}{h_b} = \frac{3.31}{h_b}$$
$$h_b = 3.68 \text{ m} \approx 3.75 \text{ m}$$

Therefore, water depth at wave breaking h_b is 3.75m and corresponding breaker wave height is 3.31m. So after 375m distance at 3.75m water depth wave breaks.

as slope is 1 in 100 so , distance from shore 1m is at water depth 0.01m similarly distance from shore 1475m is at water depth 14.75m shown table in next slide.

Distance (m)	h(m)	Distance (m)	h(m)	Distance (m)	h(m)	Distance (m)	h(m)
1	0.01	439	4.39	875	8.75	1325	13.25
25	0.25	450	4.5	900	9	1350	13.5
50	0.5	475	4.75	925	9.25	1375	13.75
75	0.75	500	5	950	9.5	1400	14
100	1	525	5.25	975	9.75	1425	14.25
125	1.25	550	5.5	1000	10	1450	14.5
150	1.5	575	5.75	1025	10.25	1475	14.75
175	1.75	600	6	1050	10.5		
200	2	625	6.25	1075	10.75		
225	2.25	650	6.5	1100	11		
250	2.5	675	6.75	1125	11.25		
275	2.75	700	7	1150	11.5		
300	3	725	7.25	1175	11.75		
325	3.25	750	7.5	1200	12		
350	3.5	775	7.75	1225	12.25		
375	3.75	800	8	1250	12.5		
400	4	825	8.25	1275	12.75		
425	4.25	850	8.5	1300	13		

So, from distance 1m to 375m , we use to Calculate setup $\bar{\eta}_{su} = \frac{h_b - h}{1 + (8/3\gamma^2)} + \bar{\eta}_{sdb}$, and

from 4m to 1475m we use to Calculate set-down $\bar{\eta}_{sd} = -\frac{1}{8} \frac{kH^2}{\sinh(2kh)}$

For water depth of $h = 14.75\text{m}$ (i.e. set down)

$$L_0 = 1.56T^2 = 1.56 \times 10^2 = 156\text{m}$$

$$\frac{h}{L_0} = \frac{14.75}{156} = 0.095$$

$$\frac{h}{L} = 0.137 \text{ (from wave table)}$$

$$L = \frac{h}{0.137} = \frac{14.75}{0.137} = 107.664\text{m}$$

$$C_0 = \frac{L_0}{T} = \frac{156}{10} = 15.6 \text{ m/s}$$

$$C = \frac{L}{T} = \frac{107.66}{10} = 10.76 \text{ m/s}$$

$$k = \frac{2\pi}{L} = \frac{2\pi}{107.664} = 0.0584$$

$$n = 0.8187 \text{ (from wave table)}$$

From shoaling formula,

$$H = H_o \times \sqrt{\frac{C_0}{2nC}} = 3 \times \sqrt{\frac{15.6}{2 \times 0.8187 \times 10.76}} = 2.822\text{m}$$

d/L_0	d/L	$2\pi d/L$	$\tanh(2\pi d/L)$	$\sinh(2\pi d/L)$	$\cosh(2\pi d/L)$	H/H_0	K	$4\pi d/L$	$\sinh(4\pi d/L)$	$\cosh(4\pi d/L)$	n	C_g/C_0	M
0.0910	0.1331	0.8362	0.6838	0.9371	1.3705	0.9411	0.7297	1.6724	2.5685	2.7563	0.8256	0.5645	10.55
0.0920	0.1340	0.8417	0.6867	0.9447	1.3757	0.9401	0.7269	1.6835	2.5993	2.7850	0.8238	0.5658	10.46
0.0930	0.1349	0.8473	0.6897	0.9524	1.3810	0.9391	0.7241	1.6946	2.6304	2.8140	0.8221	0.5670	10.38
0.0940	0.1357	0.8528	0.6925	0.9600	1.3862	0.9381	0.7214	1.7057	2.6617	2.8433	0.8204	0.5682	10.29
0.0950	0.1366	0.8584	0.6954	0.9677	1.3916	0.9371	0.7186	1.7167	2.6932	2.8729	0.8187	0.5693	10.20
0.0960	0.1375	0.8639	0.6982	0.9754	1.3969	0.9362	0.7159	1.7277	2.7251	2.9028	0.8170	0.5705	10.12
0.0970	0.1384	0.8694	0.7011	0.9831	1.4023	0.9353	0.7131	1.7387	2.7572	2.9329	0.8153	0.5716	10.04
0.0980	0.1392	0.8749	0.7038	0.9908	1.4077	0.9344	0.7104	1.7497	2.7896	2.9634	0.8136	0.5727	9.96
0.0990	0.1401	0.8803	0.7066	0.9985	1.4132	0.9336	0.7076	1.7607	2.8222	2.9941	0.8119	0.5737	9.88
0.1000	0.1410	0.8858	0.7093	1.0063	1.4187	0.9327	0.7049	1.7716	2.8551	3.0252	0.8103	0.5747	9.81
0.1010	0.1419	0.8913	0.7120	1.0140	1.4242	0.9319	0.7022	1.7825	2.8884	3.0566	0.8086	0.5757	9.73
0.1020	0.1427	0.8967	0.7147	1.0218	1.4297	0.9311	0.6994	1.7934	2.9219	3.0883	0.8069	0.5767	9.66
0.1030	0.1436	0.9022	0.7174	1.0296	1.4353	0.9304	0.6967	1.8043	2.9557	3.1202	0.8052	0.5776	9.59
0.1040	0.1444	0.9076	0.7200	1.0374	1.4409	0.9296	0.6940	1.8152	2.9897	3.1526	0.8036	0.5786	9.52
0.1050	0.1453	0.9130	0.7226	1.0453	1.4466	0.9289	0.6913	1.8261	3.0241	3.1852	0.8019	0.5794	9.45
0.1060	0.1462	0.9184	0.7252	1.0531	1.4523	0.9282	0.6886	1.8369	3.0588	3.2181	0.8003	0.5803	9.38
0.1070	0.1470	0.9239	0.7277	1.0610	1.4580	0.9275	0.6859	1.8477	3.0938	3.2514	0.7986	0.5812	9.32
0.1080	0.1479	0.9293	0.7302	1.0689	1.4637	0.9269	0.6832	1.8585	3.1291	3.2850	0.7970	0.5820	9.25
0.1090	0.1488	0.9346	0.7328	1.0768	1.4695	0.9263	0.6805	1.8693	3.1648	3.3190	0.7953	0.5828	9.19
0.1100	0.1496	0.9400	0.7352	1.0847	1.4753	0.9256	0.6778	1.8801	3.2007	3.3533	0.7937	0.5836	9.13

Dr. N

Calculation of set-down at 14.75 m

$$\bar{\eta}_{sd} = -\frac{1}{8} \frac{kH^2}{\sinh(2kh)} = -\frac{1}{8} \frac{0.0584 \times 2.822^2}{\sinh(2 \times 0.0584 \times 14.75)} = -0.0214\text{m}$$

For water depth of $h = 0.01\text{m}$ (i.e. set up)

$$\bar{\eta}_{su} = \frac{h_b - h}{1 + (8/3\gamma^2)} = \frac{3.75 - 0.01}{1 + (8/3 \times 0.9^2)} = 0.8713\text{m}$$

Dr. Mohammad Saud Afzal

Distance (m)	Setup (m)	Distance (m)	Set down (m)	Distance (m)	Set down (m)	Distance (m)	Set down (m)
1	0.8713	439	-0.1476	875	-0.0503	1325	-0.0256
25	0.817	450	-0.1419	900	-0.0481	1350	-0.0249
50	0.7588	475	-0.1305	925	-0.046	1375	-0.0241
75	0.7005	500	-0.1206	950	-0.0441	1400	-0.0234
100	0.6423	525	-0.1119	975	-0.0423	1425	-0.0227
125	0.584	550	-0.1041	1000	-0.0406	1450	-0.022
150	0.5258	575	-0.0972	1025	-0.039	1475	-0.0214
175	0.4675	600	-0.0909	1050	-0.0376		
200	0.4093	625	-0.0853	1075	-0.0361		
225	0.351	650	-0.0803	1100	-0.0348		
250	0.2928	675	-0.0757	1125	-0.0336		
275	0.2346	700	-0.0715	1150	-0.0324		
300	0.1763	725	-0.0677	1175	-0.0313		
325	0.1181	750	-0.0642	1200	-0.0302		
350	0.0598	775	-0.0609	1225	-0.0292		
375	0.0016	800	-0.058	1250	-0.0282		
400	-0.0567	825	-0.0552	1275	-0.0273		
425	-0.1149	850	-0.0527	1300	-0.0265		

Variation in wave setdown and setup

